DATA 260 Homework 5

Rental Housing Listings Extension, MCP Tools, and Safe Agent Harness

GitHub repository: https://github.com/akankshashuklaWork/data260-7838

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Hardware: MacBook Pro with Apple M4 Pro and 48 GB RAM. Local model: qwen2:7b through Ollama. Submission commit is recorded by reports/hw05/verification.json and should be replaced with the final tagged commit hash after tagging hw5.

Configuration

Value	Result
SID4	7838
PORT_BASE	8638
PREFIX	s7838
SEED	7838
VERIFY_SEED	267838
DOMAIN_ID	6

Part 1 Database and API

The existing rental-housing service was extended in code/hw4_backend rather than copied into a new service. PropertyOwner is the related entity. RentalHousingListing now has a unique listing_code, available_units with a default of one, owner_id, and created_at/updated_at timestamps. The owner foreign key uses RESTRICT behavior, and the API returns HTTP 409 when an owner still has listings.

CRUD routes cover /owners and /listings with pagination, individual lookup, update, delete, Pydantic validation, unique-field conflict handling, and /owners/{owner_id}/listings for the relationship query.

Part 1 Redux Client

The React data layer now uses Redux Toolkit in code/hw4_frontend/src/features/listings/listingsSlice.js and src/store/store.js. Axios thunks implement fetch, create, update, and delete; the Home screen reads from Redux state and updates after each successful mutation. Create and Update forms include owner selection and the unique listing code.

Part 2 MCP Servers

code/hw5_mcp/meals_server.py defines four TheMealDB tools and sends all STDIO logging to stderr. code/hw5_mcp/domain_server.py exposes exactly three domain tools: search, detail, and aggregate. All domain responses reuse the {ok, data, error} envelope, and the SQLAlchemy adapter can read the existing s7838_rel database while offline tests use deterministic fixtures.

Part 3 Retry Metrics

Failure rate	Calls	Success	Mean ms	p99 ms
0%	50	100.00%	3.500	3.500
20%	50	100.00%	3.770	5.000
50%	50	94.00%	4.850	8.000

The raw 150 records are in reports/hw05/raw/fault_injection_records.json. The experiment uses VERIFY_SEED 267838, a maximum of three retries, and bounded exponential backoff. The deterministic latency metric is documented in METRICS.md.

Part 4 Offline Tests and Part 5 Agent

code/run_hw5_tests.py reports 9/9 passing tests, including valid and invalid inputs for all three domain tools, the safety-rule allow/block pair, and the max_steps MockModel test. execute_tool never raises to its caller and always returns a JSON string. The agent records tool calls, inputs, results, stop reasons, and summaries in reports/hw05/raw/agent_runs.jsonl.

Evidence status

Completed machine-readable evidence includes verification.json, RUN_LOG.txt, METRICS.md, AI_USE.md, REFLECTION.md, raw retry records, MCP-style domain outputs, and the offline agent log. Live MCP Inspector screenshots and four Ollama scenario screenshots require the local mcp[cli] package, a running MCP Inspector, and a running Ollama service; those runtime-dependent artifacts are intentionally not represented as completed here.